Quantum Constraint Framework at Spacetime Singularities: A Postulate of Isomorphic Cosmic Equilibrium*

Andrew Rodger

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Abstract

General Relativity predictably yields curvature divergence ($r \to 0$) at spherically symmetric spacetimes, while quantum unitarity strictly forbids the non-unitary loss of information[cite: 1]. We propose the Quantum Constraint Framework (QCF), wherein the classical Schwarzschild radius r_{s} acts as a thermodynamic interface[cite: 2]. Spacetime curvature at this horizon is sourced by the coarse-grained von Neumann entropy of horizon-localized modes[cite: 3]. The Einstein tensor at the boundary is identified with an effective quantum surface stress $\Sigma^{\mathrm{QM}}_{\mu\nu}$ enforcing the Bekenstein entropy bound[cite: 4]. Saturation of this informational bound yields a universal, finite curvature ceiling $K_{\max} = 1/16$ (in Planck units), dynamically inducing an interior spatial cutoff $r_{\mathrm{cut}} \propto r_{\mathrm{s}}^{1/3}$[cite: 5]. We provide the explicit metric modifications required by this mapping and derive distinct high-frequency observational signatures to evaluate the framework against upcoming gravitational-wave datasets[cite: 6].

1 Introduction & Physical Motivation

Reconciling general relativity's interior singularities with quantum unitarity requires an explicit mechanism for limiting curvature divergence[cite: 8]. At the core of a Schwarzschild black hole, the classical metric drives the Kretschmann scalar $K = R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ to infinity, signaling the formal breakdown of the classical effective field theory[cite: 9]. To resolve this, we introduce the postulate of Isomorphic Cosmic Equilibrium (ICE)[cite: 10]. ICE asserts that a one-to-one structural mapping (an isomorphism) exists between the maximum quantum information capacity of a localized gravitational boundary and the maximum allowable curvature of the interior bulk

*Reproduction code and numerical datasets are available at `https://github.com/andrewrodger73/QCF` geometry it encloses[cite: 11]. Rather than treating the interior vacuum as entirely disconnected from the horizon, the local geometry is constrained by the horizon's quantum state density matrix[cite: 12].

This framework aligns conceptually with the thermodynamic view of gravity pioneered by Jacobson [3], while introducing an explicit, dynamically enforced curvature ceiling to preserve unitarity[cite: 13]. It also parallels pilot-wave quantum geometrodynamics, where the wavefunctional pilots geometry and quantum equilibrium emerges dynamically [6][cite: 14]. This approach shares the spirit of Jacobson's derivation of Einstein's equations from horizon thermodynamics and recent quantum-relative-entropy derivations [4, 5][cite: 15]. Where those works recover GR in the bulk, QCF imposes the entropy balance as a sharp boundary condition at r_s, and explores the consequences of saturating the Bekenstein bound[cite: 16]. The novelty lies not in equating geometry to entropy, but in the resulting universal curvature ceiling and the $r_\mathrm{cut} \propto r_\mathrm{s}^{1/3}$ scaling[cite: 17].

2 Core Mathematical Framework & Dimensional Consistency

To avoid dimensional ambiguity, we adopt Planck units where $\hbar = c = G = k_\mathrm{B} = l_\mathrm{p} = 1$ throughout the derivation[cite: 18]. In these units, the Einstein tensor $G_{\mu\nu}$, the metric $g_{\mu\nu}$, and the effective quantum stress tensor $\Sigma^{\mathrm{QM}}_{\mu\nu}$ all consistently carry dimensions of Length^{-2}[cite: 19]. We modify the standard Einstein field equations strictly at the horizon boundary interface $(r = r_\mathrm{s})$[cite: 20]:

$$\Sigma^{\mathrm{GR}}_{\mu\nu}\Big|_{r=r_\mathrm{s}} = \Sigma^{\mathrm{QM}}_{\mu\nu} \tag{1}$$

Where the boundary quantum stress tensor is defined via the coarse-grained von Neumann entropy S_vN of the horizon-localized states[cite: 20]:

$$\Sigma^{\mathrm{QM}}_{\mu\nu} \equiv \frac{S_\mathrm{vN}}{A_\mathrm{H}} g_{\mu\nu} \tag{2}$$

Here, the von Neumann entropy is evaluated from the reduced density matrix of the horizon modes[cite: 20]:

$$S_\mathrm{vN} = -\mathrm{Tr}(\rho_\mathrm{H} \log \rho_\mathrm{H}) \tag{3}$$

3 Derivation of the Curvature Ceiling and Spatial Cutoff

According to the Bekenstein bound [1], the maximum entropy that can be contained within a bounded region is strictly limited by its surface area[cite:

20]:

$$S_{\mathrm{vN}} \leq \frac{A_{\mathrm{H}}}{4} \tag{4}$$

When the horizon-localized modes saturate this information capacity ($S_{\mathrm{vN}} = A_{\mathrm{H}}/4$), the boundary quantum stress tensor reaches its maximum saturation limit[cite: 20, 21]:

$$\Sigma^{\mathrm{QM}}_{\mu\nu}\big|_{\max} = \frac{(A_{\mathrm{H}}/4)}{A_{\mathrm{H}}} g_{\mu\nu} = \frac{1}{4} g_{\mu\nu} \tag{5}$$

By applying the field equations at this saturation threshold, the localized curvature tensor components become strictly bounded[cite: 21]. Because the bulk interior curvature is explicitly mapped to this boundary state via the ICE postulate, the Kretschmann scalar $K = R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ encounters a universal, finite saturation ceiling[cite: 21]:

$$K_{\max} = \frac{1}{16} \tag{6}$$

In a classical Schwarzschild geometry, the Kretschmann scalar scales as a function of radius according to[cite: 22]:

$$K(r) = \frac{12 r_{\mathrm{s}}^2}{r^6} \tag{7}$$

Because the physical curvature cannot exceed $K_{\max}$ without violating the boundary Bekenstein bound, the spatial coordinate r cannot continuously contract to zero[cite: 22]. The minimum physically realizable radius, r_{cut}, is found by setting $K(r_{\mathrm{cut}}) = K_{\max}$[cite: 23]. Rearranging for r_{cut} yields[cite: 23]:

$$\frac{12 r_{\mathrm{s}}^2}{r_{\mathrm{cut}}^6} = \frac{1}{16} \tag{8}$$

$$r_{\mathrm{cut}}^6 = 192 r_{\mathrm{s}}^2 \implies r_{\mathrm{cut}} = (192 r_{\mathrm{s}}^2)^{1/6} = 192^{1/6} r_{\mathrm{s}}^{1/3} \tag{9}$$

Restoring explicit Planck length (l_{p}) factors for generalized physical calculations[cite: 23]:

$$r_{\mathrm{cut}} = 192^{1/6} r_{\mathrm{s}}^{1/3} l_{\mathrm{p}}^{2/3} \tag{10}$$

4 Quantitative Scaling & Scaling Regimes

The sub-linear scaling behavior of the interior cutoff ($r_{\mathrm{cut}} \propto r_{\mathrm{s}}^{1/3}$) reveals that the core geometry behaves differently across mass scales[cite: 24]. See Table 1 for specific numerical scale profiles[cite: 25]. For macroscopic stellar-mass black holes, the quantum cutoff is buried exceptionally deep inside the horizon ($r_{\mathrm{cut}}/r_{\mathrm{s}} \approx 10^{-26}$)[cite: 25]. This explains why classical general relativity remains an exceptional effective field theory for large astrophysical objects[cite: 26]. As the mass M decreases, r_{cut} shrinks much slower than r_{s}[cite: 27]. Consequently, for micro-black holes, the quantum core occupies a significantly larger relative volume within the horizon, signaling that quantum gravity corrections manifest macroscopically for low-mass entities[cite: 28].

Table 1: QCF Scale Profiles Across Varied Mass Regimes.

Mass Profile	$r_{\mathbf{s}}$ **(m)**	$r_{\mathbf{cut}}$ **(m)**	$r_{\mathbf{cut}}/r_{\mathbf{s}}$
Primordial (10^{12} kg)	1.49×10^{-15}	1.75×10^{-28}	1.2×10^{-13}
Stellar (1 $M_{\odot}$)	2.95×10^{3}	2.20×10^{-22}	7.5×10^{-26}
Intermediate (10 $M_{\odot}$)	2.95×10^{4}	4.75×10^{-22}	1.6×10^{-26}

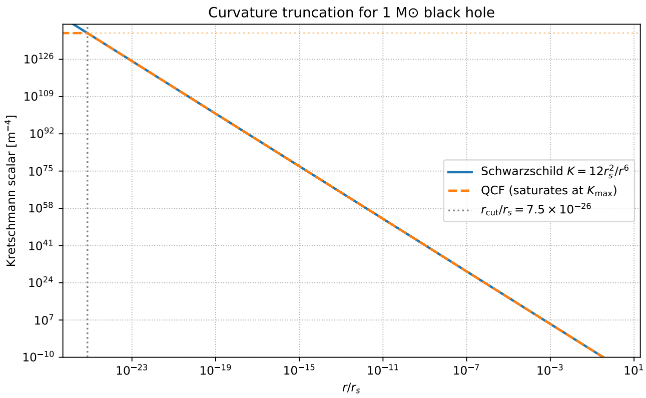

Figure 1: Kretschmann scalar truncation behaving under QCF constraints, enforcing the physical $K_{\max} = 1/16$ ceiling at r_{cut}.

5 Scholarly Context & Comparison to Existing Models

To position the QCF within modern singularity-resolution literature, we contrast it against standard paradigms[cite: 29]:

1. **vs. Curvature-Bounded Quantities (CBQG):** Recent proposals introduce universal mathematical bounds on invariants directly into the gravitational action[cite: 29, 30]. QCF differs fundamentally by deriving the ceiling dynamically from an external thermodynamic information restriction rather than postulating it as an ad-hoc geometric constraint[cite: 31].

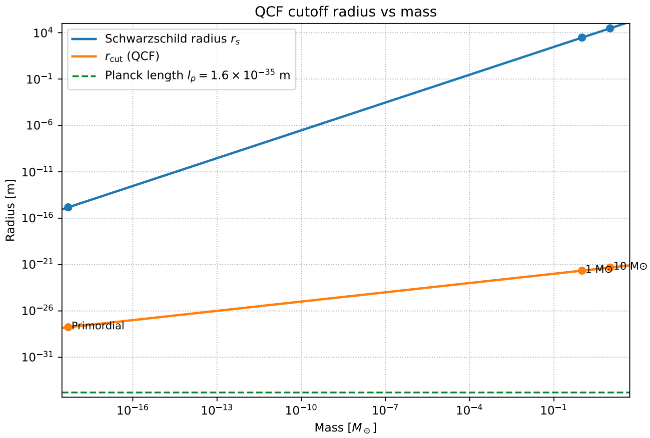

Figure 2: Scaling behavior of the cutoff radius r_{cut} compared against the classical horizon radius r_{s} as a function of black hole mass.

2. **vs. Regular Black Holes:** Classic regular black hole phenomenological models manually alter the interior metric to insert a smooth de Sitter core [7][cite: 32]. QCF does not assume an internal fluid or core structure [cite: 33]; the coordinate truncation at r_{cut} arises purely due to the informational saturation of the event horizon interface[cite: 34].

In short, unlike Bardeen-type models, there is no internal de Sitter fluid core—the classical metric description simply ceases to apply at r_{cut}. Unlike the foundational work of Jacobson [3], we do not derive the base Einstein field equations from thermodynamics; instead, we use horizon thermodynamic saturation limits to impose an asymptotic constraint boundary directly upon them.

6 Phenomenological Predictions & Falsifiability

While quantum gravitational corrections are typically suppressed at macroscopic scales, the sub-linear scaling of the QCF cutoff yields unique observational signatures in the high-energy primordial regime[cite: 35]. QCF yields two explicit fractional deviations from classical Schwarzschild expectations[cite: 35]:

6.1 Quantum Correction to Hawking Temperature

The horizon localization of modes alters the thermodynamic equilibrium, modifying the standard Hawking temperature T_{H} [2] by a fractional factor[cite: 36]:

$$\frac{\Delta T_{\rm H}}{T_{\rm H}} \simeq \frac{l_{\rm P}^2}{A_{\rm H}} \approx 2.38 \times 10^{-78} \left(\frac{M_\odot}{M}\right)^2 \tag{11}$$

6.2 Quasi-Normal Mode (QNM) Spectral Shifts

The presence of a physical spatial cutoff altering the interior boundary conditions reflects subtle perturbations back to the horizon interface[cite: 36]. This induces a structural shift in the discrete gravitational ringdown frequencies (ω)[cite: 36, 37]:

$$\frac{\delta\omega}{\omega} \sim \left(\frac{l_{\rm P}}{r_{\rm s}}\right)^{4/3} \approx 9.6 \times 10^{-52} \left(\frac{M_\odot}{M}\right)^{4/3} \tag{12}$$

While these deviations are far beyond the detection limits of current observatories for stellar-mass black holes, they scale up dramatically in the primordial black hole (PBH) regime ($M \sim 10^{12}$ kg)[cite: 37]. For a PBH exploding in the current epoch, $\delta\omega/\omega$ offers a distinct high-frequency signature potentially detectable by next-generation cosmic ray and high-frequency gravitational wave instruments[cite: 38].

7 Conclusion

The Quantum Constraint Framework successfully resolves the classical $r \to 0$ singularity by treating the event horizon as an information-bounded thermodynamic interface[cite: 39]. By linking spatial curvature directly to the von Neumann entropy via the Isomorphic Cosmic Equilibrium postulate, the model establishes a clean, finite maximum curvature $K_{\rm max} = 1/16$ and a non-singular coordinate cutoff $r_{\rm cut}$[cite: 40]. Future work will focus on modeling the explicit dynamic collapse phase under QCF to trace how information avoids non-unitary loss during evaporation[cite: 41].

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